Summary

Where the Project Stands

Short version — see WorkPlan.tex for derivations

Step 1 — Discrete objective

$$U[\mathbf{x}] = \sum_{j=1}^{M} \left[\tfrac{1}{2}\gamma_j X_j^2 + \varepsilon_j \sum_{k=1}^{N} |n_{jk}| + \frac{\tilde{\eta}_j}{\tau} \sum_{k=1}^{N} n_{jk}^2 \right] + \lambda \sum_{k=1}^{N} \tau\, \mathbf{x}_k^\top C \mathbf{x}_k \tag{1}$$

How we got here: taken directly from Almgren–Chriss (2000) [1] — Eq. 15 ($U = E + \lambda V$) applied to their Eqs. 29–30 (expected cost, variance), restricted to diagonal impact matrices (their Eq. 32).

Step 2 — Continuum running-cost integral

$$J[\mathbf{v}] = \int_0^T \left(\mathbf{v}(t)^\top H \mathbf{v}(t) + \lambda\, \mathbf{x}(t)^\top C \mathbf{x}(t) \right) dt, \qquad \dot{\mathbf{x}} = -\mathbf{v}, \qquad \mathbf{x}(0) = \mathbf{X},\ \mathbf{x}(T) = \mathbf{0} \tag{2}$$

How we got here: start from Step 1, drop the two terms that don't depend on the schedule (permanent impact, spread cost under sales-only), write $n_{jk} = \tau v_j(t_k)$, and take $\tau \to 0$: the Riemann sum $\sum_k \tau f(t_k)$ becomes the integral $\int_0^T f(t)\, dt$. This is the multi-asset continuum limit of Bergault–Drissi–Guéant (2022) [2]; the single-asset case matches Almgren–Chriss (2000) [1].

Step 3 — HJB equation and matrix Riccati ODE

$$\partial_t \mathcal{V} - \tfrac{1}{4} (\nabla_{\mathbf{x}} \mathcal{V})^\top H^{-1} (\nabla_{\mathbf{x}} \mathcal{V}) + \lambda\, \mathbf{x}^\top C \mathbf{x} = 0, \qquad \dot{P} = \tfrac{1}{2} P H^{-1} P - 2\lambda C, \qquad P(T) = \Phi \tag{3}$$

How we got here: define the cost-to-go $\mathcal{V}(\mathbf{x}, t) = \min_{\mathbf{v}} \int_t^T (\cdots)\, ds$ on Step 2's integral, apply Bellman's dynamic programming principle (split the horizon into $[t, t+dt] + [t+dt, T]$, expand to first order) to get the HJB PDE, then substitute the quadratic ansatz $\mathcal{V} = \frac{1}{2}\mathbf{x}^\top P(t) \mathbf{x}$ to collapse it to the Riccati ODE. Verified by hand against Bergault–Drissi–Guéant (2022) [2], Eqs. 13–14 and 20–21.

Reading the status column. Three different kinds of claim get mixed together if not kept separate, so each row is tagged with which one applies:

- *Numerically validated* — an actual number was computed (by hand or by script) and compared against another number. This is the only kind of "done" that means the formula was tested, not just read.
- *Literature-matched* — the AI assistant re-derived the algebra this session and confirmed it agrees symbol-for-symbol with a cited published paper. This means the formula is traceable to a real source, *not* that it has been independently tested by a human or against your own numbers.
- *Derived, unchecked* — written down (by this project or the AI) but never compared against anything.

Stage	Status	Note
1. Discrete objective	Sourced; doc repairs open	Formula taken directly from Almgren–Chriss (2000) [1]. **Not fully done**: Eq. 20, Eqs. 26–27, Eq. 32 (H vs. $\tilde{H}$), and the unfinished Drift sentence are still open repairs in the reference document (see WorkPlan Stage 1).
2. HJB, Riccati, trajectory ($\Phi \to \infty$)	Literature-matched + partly numerically validated	HJB/Riccati/trajectory algebra matches Bergault–Drissi–Guéant (2022) [2] and Almgren–Chriss (2000) App. A.2 [1] (AI-checked this session). Separately, *numerically* validated only for: one asset (exact), one asset vs. discrete cost at $N = 4$ ($\sim 3 \times 10^{-4}$ agreement), and two assets for **one** parameter set. Not a general proof.
2 (finite Φ)	Literature-matched only	Closed form in Bergault et al. [2] App. A, Cor. 1. AI-checked against the paper; **not yet numerically tested** against anything in this project.
2b. Time-dependent $\lambda(t), H(t), C(t)$	Literature-matched (Riccati form only)	Drissi (2023) [3] confirmed real and on-topic (AI-checked). Closed form exists only in the special volume-time-clock case [4] (derived here, standard calculus, **not numerically checked**). General case: no closed form, not implemented.
3. Concrete instance	In progress, own work	No paper needed — this is a parameter choice, not a result to cite. Known failure mode already observed: high correlation + small second position breaks the sales-only assumption.
4. Cole–Hopf linearisation	Derived, unchecked	This project's own instantiation of a standard technique (Fleming–Soner [5], Kappen [6]); **no numerical check has been run**.
5. Oscillator spectrum / choose ν	Derived, unchecked	Own work; the ν-sweep table (exit criterion) has not been produced.
6. Discretisation	Not started	
7. Schrödingerisation	Not started	Jin–Liu–Yu papers [7, 8] not yet read; current notes are reconstructed from memory, explicitly marked unreliable in WorkPlan.
8. Resource count (N_q, N_T)	Not started	
9. Readout cost	Not started	
10. Sweeps & write-up	Not started	

Next step

The single blocking action is **Stage 3**: pick one concrete numeric instance $(\eta, \sigma, \rho, \lambda, T, X)$ and check, for that instance only: $C \succeq 0$, $\eta_j > 0$, real κ_i, and $v_j(t) \geq 0$ for all t (the sales-only check — known to fail at high correlation with an asymmetric position, so this is not a rubber stamp). Nothing in Stages 4–9 can produce an actual number before this is fixed; right now they only have symbols.

Two things should happen alongside it, not instead of it:

- Close the Stage 1 reference-document repairs (Eq. 20, Eqs. 26–27, Eq. 32, the Drift sentence) in Almgren–Chriss [1] — small, independent of Stage 3.

- Before trusting any Stage 7–9 number later, read the Jin–Liu–Yu Schrödingerisation papers [7, 8]; everything there is currently reconstructed from memory, not sourced.

Parameter glossary (Stage 3)

Symbol	Meaning
η_j	Temporary market impact of asset j; cost of trading fast. Larger η_j penalises quick selling more.
σ_j	Volatility of asset j's price, per unit time.
ρ	Correlation between the two assets' returns; builds the covariance matrix $C = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$.
λ	Risk aversion. Larger λ makes holding inventory costlier, pushes faster liquidation.
T	Horizon: time allowed to fully liquidate.
X	Starting inventory vector (shares held per asset) at $t = 0$.
$\kappa_i = \sqrt{\lambda\mu_i}$	Decay rate of each decoupled trading mode (after diagonalising $H^{-1/2}CH^{-1/2}$ [1, 2]); sets how fast that mode liquidates. μ_i are the eigenvalues of $H^{-1/2}CH^{-1/2}$.
$v_j(t) = -\dot{x}_j(t)$	Selling rate of asset j at time t. Must stay ≥ 0 everywhere, or the sales-only assumption breaks (the model has no cost term for buying back).

References

[1] R. Almgren and N. Chriss, "Optimal Execution of Portfolio Transactions," *J. Risk* **3**(2), 5–39 (2000).

[2] P. Bergault, F. Drissi, and O. Guéant, "Multi-asset Optimal Execution and Statistical Arbitrage Strategies under Ornstein–Uhlenbeck Dynamics," *SIAM J. Financial Math.* **13**(1), 353–390 (2022); arXiv:2103.13773.

[3] F. Drissi, "Solvability of Differential Riccati Equations and Applications to Algorithmic Trading with Signals," *Appl. Math. Finance* **29**(6), 457–493 (2023), arXiv:2202.07478.

[4] O. Guéant, *The Financial Mathematics of Market Liquidity: From Optimal Execution to Market Making*, Chapman and Hall/CRC (2016).

[5] W. H. Fleming and H. M. Soner, *Controlled Markov Processes and Viscosity Solutions*, 2nd ed., Springer (2006).

[6] H. J. Kappen, "Linear theory for control of nonlinear stochastic systems," *Phys. Rev. Lett.* **95**, 200201 (2005).

[7] S. Jin, N. Liu, and Y. Yu, "Quantum simulation of partial differential equations via Schrödingerization," *Phys. Rev. Lett.* **133**, 230602 (2024).

[8] S. Jin, N. Liu, and Y. Yu, "Quantum simulation of partial differential equations via Schrödingerisation: technical details," *Phys. Rev. A* **108**, 032603 (2023), arXiv:2212.14703.